

XDL-energy: Explainable hybrid deep learning framework for smart campus energy forecasting and anomaly justification using the DPU-ALDOSKI dataset

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ABSTRACT

Managing energy efficiently and in a way that is understandable is essential for the sustainable and intelligent operation of modern smart campuses. In this paper, we introduce XDL-Energy, a novel framework that combines a hybrid Convolutional Neural Network–Long Short-Term Memory (CNN-LSTM) model with SHapley Additive exPlanations (SHAP) to deliver both high-accuracy energy consumption forecasting and interpretable anomaly detection. The model is trained on the real-world DPU-ALDOSKI dataset, which contains over 3 million multi-sensor IoT records from distributed campus buildings. By fusing CNN's spatial pattern recognition with LSTM's temporal sequence modeling, the framework captures complex energy behaviors across time and space. SHAP analysis reveals key drivers of predictions, such as temperature, motion, and humidity, offering both global and local interpretability. Experimental results demonstrate that XDL-Energy achieves robust performance (MAE: 0.045, RMSE: 0.062, MAPE: 4.8%) and reliably flag anomalies using dynamic rolling thresholds. Additional evaluations including ablation studies, SHAP dependence plots, and multiple anomaly case analyses confirm the system's practical relevance. XDL-Energy bridges the gap between deep learning and transparency, making it a deployable and trustworthy solution for smart campus energy management.

1. Introduction

Over the past decade, the rise of smart campuses enabled by the Internet of Things (IoT), cloud platforms, and embedded sensors has transformed traditional energy systems into intelligent, adaptive infrastructures [1,2]. These IoT-driven campuses now collect vast amounts of real-time data on energy consumption, environmental conditions, and operational behavior. In such environments, efficient energy management is not merely an option; it is a necessity to reduce operational costs, promote sustainability, and ensure the resilience of institutional systems.

While large-scale data collection is a step forward, the real challenge lies in making sense of this data to inform timely decisions. Traditional rule-based systems are no longer adequate in handling the complexity and dynamic nature of campus energy usage. As a result, machine learning (ML) and more recently, deep learning (DL) models have become prominent tools for energy prediction and anomaly detection [3,4]. These models, particularly architectures like LSTM and CNN, are well-suited for time-series forecasting and feature extraction, respectively [5,6]. They can uncover hidden trends, seasonality, and

correlations among sensor inputs that would otherwise be missed. However, despite their predictive power, these models often function as “black boxes,” offering little to no insight into why certain predictions are made a major limitation in high-stakes domains like energy infrastructure [7].

The lack of interpretability in deep learning models introduces trust and accountability issues. Campus facility managers and engineers need to understand not only what is predicted but why. For instance, when an anomaly in energy consumption is detected, understanding whether it is due to external temperature, faulty equipment, or unusual occupancy patterns is crucial for informed response. This has led to growing interest in Explainable Artificial Intelligence (XAI), which aims to make machine learning models more transparent and understandable to human users [8,9].

In response to this need, we propose XDL-Energy, an explainable hybrid deep learning framework designed for energy forecasting and anomaly explanation in smart campus environments. At the core of XDL-Energy is hybrid CNN-LSTM architecture, which leverages the spatial learning capabilities of Convolutional Neural Networks (CNNs) and the temporal sequence modeling of Long Short-Term Memory (LSTM)

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networks [10,11]. This architecture is uniquely suited to the DPU-ALDOSKI dataset, a rich collection of real-time, multi-sensor energy data from an IoT-based campus infrastructure [12]. The hybrid model is capable of accurately forecasting energy consumption trends while simultaneously flagging deviations from expected behavior.

To address the black-box nature of deep learning, we incorporate Shapley Additive explanations (SHAP) into the framework. SHAP is a model-agnostic tool from the XAI family that quantifies the contribution of each input feature to a given model prediction [13]. In the context of XDL-Energy, SHAP helps uncover which factors such as temperature, humidity, motion sensors, or occupancy play the most significant role in both normal energy forecasts and detected anomalies. This level of interpretability turns a deep model from a prediction tool into a decision support system that is both powerful and accountable.

The DPU-ALDOSKI dataset, developed at Duhok Polytechnic University, serves as the foundation for training and evaluating XDL-Energy. It includes millions of energy consumption records collected from distributed IoT devices across various facilities and departments [12]. The dataset contains not only energy readings, but also contextual information such as temperature, humidity, motion detection, and time-stamped occupancy indicators. Such granularity enables the model to develop rich temporal and contextual associations, improving both prediction accuracy and anomaly explanation.

Our experimental results reveal that the XDL-Energy framework performs exceptionally well in forecasting energy consumption with low error margins (measured by MAE and RMSE). More importantly, the SHAP component provides meaningful explanations for high-risk anomalies, allowing stakeholders to pinpoint the causes and take corrective action quickly. The explainability of the system also enables cross-functional collaboration between data scientists, facility managers, and IT departments in understanding and improving energy performance.

To summarize, the contributions of this paper are threefold:

- We develop a hybrid CNN-LSTM model that learns both spatial and temporal dynamics of energy usage in a smart campus setting.
- We integrate SHAP explanations to provide feature-level interpretations of model predictions and anomaly triggers, enhancing trust and transparency.
- We validate the framework using a real-world, large-scale IoT energy dataset (DPU-ALDOSKI), demonstrating both high predictive performance and actionable insight generation.

While CNN-LSTM models and SHAP explanations have been applied in smart home and building environments, smart campuses present distinct challenges due to their multi-zonal infrastructure, heterogeneous energy loads (e.g., labs, classrooms, dorms), and highly dynamic occupancy patterns that vary by schedule, academic calendar, and usage type. Unlike residential or office settings, campuses experience frequent operational shifts influenced by class timetables, exams, and seasonal events. To the best of our knowledge, this is the first work that combines CNN-LSTM forecasting with SHAP-based interpretability for real-time anomaly detection in multi-sensor, real-world smart campus energy data, using a dataset collected across multiple cities and academic departments. Our framework XDL-Energy not only provides robust energy forecasts but also justifies anomalies in a context-aware manner, bridging the gap between deep learning accuracy and explainability in complex academic infrastructures.

Campus energy consumption differs significantly from residential or standard commercial buildings due to its heterogeneous infrastructure, distributed IoT devices, irregular schedules, and diverse usage patterns. University campuses comprise classrooms, laboratories, dormitories, and administrative buildings, each with distinct operational profiles and varying occupancy levels throughout the day. These fluctuations, combined with frequent event-driven anomalies (e.g., exams, night lectures, laboratory experiments), introduce complex non-linear patterns that

challenge traditional forecasting models. Hence, focusing on campuses allows us to design and evaluate an energy management framework capable of handling high variability and providing actionable insights for large-scale, distributed educational environments.

In the following sections, we review related work in smart energy forecasting and explainable AI, describe our dataset and hybrid architecture in detail, present experimental results, and discuss the practical implications of XDL-Energy in smart campus deployments.

2. Related work

Energy consumption forecasting has long been studied in buildings, from classical time-series and regression baselines to today's deep-learning hybrids. Traditional models such as ARIMA and linear/multivariate regression remain useful but struggle with nonlinearity and regime shifts common in operational data from real buildings [3,5]. With rich IoT telemetry, data-driven methods especially LSTM/GRU and hybrid CNN-RNN architectures have shown clear gains for short-term load forecasting (STLF) across building types [3,6,10,14,15]. Recent studies papers continue this trend: *meta-learning* for low-data buildings [16], hybrid DL models for building-level STLF [17], and comprehensive overviews that organize methods by horizon, features, and building archetype while discussing emerging LLM-assisted workflows [18]. Classic building-scale STLF exemplars remain highly cited touchpoints for non-residential and institutional settings [19,20].

2.1. Deep learning for building-scale forecasting

Deep sequential models capture temporal dependencies and calendar/meteorological effects better than purely statistical baselines [3,10,14]. Hybrid designs leverage CNNs for local pattern extraction with LSTM/GRU for sequence modeling, improving accuracy and robustness in multivariate sensor streams [6]. Newer study proposes hybrid pipelines and sequence decomposition to stabilize training and sharpen peak tracking in building electricity time series [17]. When historical data are scarce, *meta-learning* and transfer across buildings can meaningfully reduce error and fine-tuning cost [16]. Clustering of time-series with shape preservation further helps segment heterogeneous load behaviors before model fitting [21].

2.2. Explainable AI (XAI) for energy prediction and anomaly analysis

Limited interpretability has slowed adoption of black-box models in mission-critical operation. SHAP and LIME are increasingly used to expose feature attributions for energy use intensity and load predictions, supporting trustworthy decision-making by operators and retrofit stakeholders [7,8,13,22–26]. In buildings specifically, recent studies examine XAI method selection versus explanation quality and user trust and propose frameworks to integrate XAI into automated energy management workflows [26,27]. These trends align with our approach, where SHAP explanations justify anomalies and surface the drivers (e.g., outdoor temperature, occupancy proxies) behind forecast deviations.

2.3. Campus and multi-building contexts

Campus environments combine diverse building typologies, irregular schedules, and distributed sensing, making them a rigorous testbed for forecasting and anomaly detection. Earlier studies on institutional and non-residential buildings established baseline practices for feature engineering and evaluation [19,20]. Recent advances in clustering and multi-source fusion help characterize occupant-driven variability, enabling better generalization across buildings with different uses or climates [21]. Our study contributes to this trajectory by applying an explainable hybrid DL framework to a **large, real-world campus IoT dataset** (DPU-ALDOSKI), and by coupling forecasting with anomaly justification to produce operator-actionable insights.

3. Research gaps and our contribution

Despite substantial progress, three gaps persist in the literature: (i) a paucity of studies that jointly deliver state-of-the-art forecasting accuracy and transparent anomaly justification on large, high-frequency campus datasets; (ii) limited evidence on data-scarce transfer and *meta-learning* strategies that enable cross-building generalization within institutional settings; and (iii) insufficient human-centered evaluations of explainability artefacts for facility practitioners. XDL-Energy addresses these gaps by (a) leveraging a multi-million-record campus IoT dataset, (b) coupling a hybrid deep forecaster with SHAP-based anomaly attribution explicitly linked to operational context, and (c) aligning with current study directions in hybrid deep learning, low-data adaptation, and explanation usability [16–20,25–27].

4. Dataset description

The proposed XDL-Energy framework is developed and evaluated using the DPU-ALDOSKI dataset, a real-world, large-scale, IoT-based energy consumption dataset collected from a university campus. The dataset consists of multi-sensor, high-frequency measurements recorded over several months, covering multiple operational zones such as classrooms, laboratories, administrative offices, and dormitories. Data were collected using a custom-built monitoring and control system integrated with several sensors and devices, described below [28]:

- DHT22 Sensor.
- PZEM-004 T Module.
- Solid State Relay (SSR).
- Hi-Link Power Supply (HLK-PM01).
- General Purpose, High Current, NPN Transistor Array (NTE929).
- Push-Button Switch.
- TFT LCD Module.
- ESP32 Microcontrollers.

The DPU-ALDOSKI dataset used in this study is publicly available for download. It contains over 3 million time-stamped multi-sensor records collected from a distributed IoT infrastructure deployed across Duhok Polytechnic University campuses. Researchers can access the dataset via the following DOI:

- [doi:10.1016/j.dib.2023.109455](https://doi.org/10.1016/j.dib.2023.109455)

It is also hosted on Mendeley Data at:

- <https://data.mendeley.com/datasets/75c8p9kkfd/1>

This ensures full reproducibility of the experiments conducted in this work. The dataset includes features such as energy consumption, voltage, temperature, humidity, motion, occupancy, and power factor, recorded at 30-second intervals across various buildings.

4.1. Data collection and scope

The DPU-ALDOSKI dataset was collected between February 13, 2022, and January 8, 2023, covering nearly eleven months of continuous operations. Data was gathered from multiple smart meters and embedded IoT sensors deployed across several departments and campus buildings in the cities of Duhok and Shekhan, including classrooms, laboratories, offices, and public utility spaces.

In total, the dataset contains over 3 million timestamped records, offering both breadth and depth in terms of data granularity and diversity. Each record is associated with a specific device or location, enabling spatial correlation analysis alongside temporal forecasting.

4.2. Features and sensor modalities

The dataset includes the following key features:

- Timestamp – Precise date and time of each reading (down to the second).
- Energy Consumption (kWh) – Total active energy consumed per reading.
- Voltage (V) – Line voltage measurements for each circuit/device.
- Current (A) – Electrical current drawn.
- Frequency (Hz) – Power supply frequency.
- Power Factor – The ratio of real power to apparent power.
- Temperature (°C) – Ambient room or device temperature.
- Humidity (%) – Relative humidity levels.
- Motion Sensor State – Binary indication (active/inactive) of motion detection.
- Occupancy Detection – State of room/device usage (based on motion and schedule).

These multi-modal features enable the modeling of complex interactions between environmental conditions, human presence, and energy usage behavior, making the dataset suitable not only for prediction tasks but also for root-cause analysis of anomalies.

4.3. Data frequency and granularity

One of the key strengths of the DPU-ALDOSKI dataset is its high temporal resolution. Sensor readings are logged at 1-minute intervals, allowing for detailed short-term forecasting and anomaly detection. This level of granularity supports models that rely on rapid variations in sensor input, such as CNN filters for pattern detection and LSTM cells for long-sequence learning.

Additionally, data is segmented by device ID and building location, enabling spatial disaggregation and comparative analysis across different zones of the campus. This is particularly useful for identifying localized inefficiencies or equipment-specific issues.

4.4. Data quality and preprocessing

Before training the hybrid deep learning model, several preprocessing steps were applied to the raw dataset:

- Missing value imputation using forward/backward fill and interpolation.
- Outlier removal through statistical thresholds and sensor consistency rules.
- Normalization of continuous variables to standard scales for model stability.
- Encoding of categorical variables (e.g., motion state, device ID).
- Sliding window transformation for time-series sequence modeling.

These preprocessing steps ensure that the model operates on clean, consistent data, improving its accuracy and generalizability. The DPU-ALDOSKI dataset offers a comprehensive foundation for predictive and explainable modeling in smart energy systems. Its multi-sensor structure, high-frequency resolution, and contextual diversity make it particularly suited for hybrid deep learning approaches that require both temporal depth and cross-feature interactions. By applying the XDL-Energy framework to this dataset, we demonstrate the practical feasibility of deploying interpretable AI tools for real-time energy management in academic institutions.

Final dataset sizes after filtering. From the original 3,210,860 time-stamped records at 30-second resolution, the normal-operation filters (timestamp de-duplication; completeness checks on core sensors; $\mu \pm 3\sigma$ outlier rule on Power; basic frequency sanity; and a Sun–Thu weekday rule) retained 1,286,654 rows (40.07 % of the original) for modeling.

These were partitioned chronologically into 900,657 rows for training (28.05 %), 192,998 for validation (6.01 %), and 192,999 for testing (6.01 %). In model terms, this corresponds to 900,627 / 192,968 / 192,969 windows for train/val/test, respectively (window length 30, horizon 1, stride 1).

5. Methodology

This section describes the proposed XDL-Energy framework, which integrates a hybrid deep learning model with explainability techniques to forecast energy consumption and detect anomalies in a smart campus setting. The architecture combines a Convolutional Neural Network (CNN) with a Long Short-Term Memory (LSTM) network to capture spatial and temporal dependencies, respectively. Furthermore, SHapley Additive exPlanations (SHAP) are employed to interpret both the global behavior and individual predictions of the model. The methodology also outlines the anomaly detection mechanism and evaluation metrics used to assess performance.

5.1. Hybrid CNN-LSTM architecture

The hybrid CNN-LSTM model is designed to exploit both spatial feature patterns in multi-sensor data and temporal dependencies in energy usage sequences. The input to the model is a multivariate time series window representing sequential sensor readings over a fixed time span (e.g., 30 min or 60 min), where each timestep includes multiple features: energy consumption, voltage, temperature, humidity, motion state, etc.

- **CNN Layer(s):** The initial layers are 1D convolutional layers that scan across timesteps to extract local temporal-spatial patterns. CNN acts as an automatic feature extractor, learning filters that identify fluctuations in voltage, current, or occupancy patterns that might influence energy consumption.
- **LSTM Layer(s):** The output from the CNN is fed into one or more stacked LSTM layers, which are responsible for learning long-term dependencies and trends. The LSTM captures how prior states influence future energy behavior over time.
- **Dense Output Layer:** The final layer is a fully connected (dense) layer that maps the learned temporal-spatial features to a single predicted value representing the next-step energy consumption.

The input to the model is a sliding window of multivariate time series data of length T , where each timestep t includes features such as energy consumption, voltage, temperature, humidity, and motion state:

$$X = \{x_1, x_2, \dots, x_T\}, \text{ where } x_t \in \mathbb{R}^d.$$

CNN Layer:

The CNN applies 1D convolution over the time dimension to extract short-term patterns. For an input sequence X and a kernel w of width k , the convolution output at position i is:

$$c_i = \sigma(\sum w_j \cdot x_{i+j} + b).$$

LSTM Layer:

The output from CNN is passed to an LSTM layer, which models long-term dependencies:

$$\begin{aligned} f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ \tilde{C}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \\ C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\ o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\ h_t &= o_t \odot \tanh(C_t) \end{aligned}$$

The final output h_T is passed through a dense layer to generate the forecast $\hat{y}_{T+1}:\hat{y}_{T+1} = W_y \cdot h_T + b_y$.

5.2. Anomaly detection mechanism

To detect anomalous energy usage patterns, we adopt a prediction-error-based thresholding approach, which is practical for real-time deployment in smart campus systems. The hybrid CNN-LSTM model is first trained exclusively on data representing normal operational behavior, allowing it to learn expected patterns under typical conditions. Normal data was selected based on a combination of statistical, temporal, and sensor consistency filters. Specifically:

- **Statistical Filtering:** Time windows where energy consumption exceeded the rolling mean plus 3 standard deviations ($\mu + 3\sigma$) over 24-hour intervals were excluded. Sensor outliers (e.g., voltage < 190 V or > 250 V; temperature $> 45^\circ\text{C}$) were also removed.
- **Temporal Heuristics:** Only data from regular working days (Sunday to Thursday) was used, excluding weekends, exam periods, and known holidays.
- **Sensor Consistency Check:** Periods with anomalous sensor behavior such as high energy during unoccupied states or inconsistent motion-occupancy alignment were removed to ensure the training data reflected stable patterns.

Once trained on this filtered data, the model's absolute prediction error at each timestep is computed as:

$$e_t = |y_t - \hat{y}_t|.$$

A dynamic threshold is then calculated using a sliding window of recent prediction errors:

$$\text{Threshold}_t = \mu_t + \alpha \cdot \sigma_t.$$

where μ_t and σ_t represent the mean and standard deviation of errors within the window, and α is a scaling factor (set to 3 by default). If $e_t > \text{Threshold}_t$ the instance is flagged as anomalous.

This method is adaptive, computationally lightweight, and suitable for online detection, making it ideal for continuous monitoring across distributed campus zones. We clarify that while this anomaly detection technique is standard, it serves here as a reliable baseline integrated with SHAP-based interpretability to support root-cause analysis and actionable decision-making.

We trained on 900,657 rows (28.05 % of the original dataset), validated on 192,998 (6.01 %), and tested on 192,999 (6.01 %) after applying the normal-operation filters in Sec. 3.4; in sequence form this equals 900,627 / 192,968 / 192,969 windows (length 30, horizon 1, stride 1).

5.3. SHAP explainability integration

To overcome the black-box nature of the hybrid CNN-LSTM model, SHapley Additive exPlanations (SHAP) are employed for post-hoc interpretability. SHAP provides both global and local explanations:

- **Global SHAP Values:** Aggregated SHAP values over the test dataset reveal which features (e.g., temperature, motion, voltage) are generally most influential in the model's decision-making. This helps understand feature importance trends across the entire system.
- **Local SHAP Explanations:** For each individual prediction especially anomalies SHAP assigns contribution values to each input feature, indicating how much each feature pushed the prediction higher or lower. For example, a spike in temperature may be shown to contribute positively to an anomaly flag, providing human-readable insight into the cause.

Visual tools such as SHAP summary plots, force plots, and dependence plots are used to present these interpretations to stakeholders, including energy managers and technical staff. To interpret predictions and anomalies, we use SHAP (SHapley Additive exPlanations). For any model prediction $f(x)$, SHAP explains it by computing feature contributions:

$$f(x) = \varphi_0 + \sum \varphi_i$$

where φ_0 is the base value (expected model output), and φ_i is the contribution of feature i . SHAP provides both global and local explanations to highlight key factors in predictions.

5.4. Model training and evaluation

The model was implemented using TensorFlow/Keras and trained on the DPU-ALDOSKI dataset using a sliding window approach for sequence generation. The dataset was split chronologically into 70 % training, 15 % validation, and 15 % test sets to preserve temporal dependencies and prevent data leakage.

Hyperparameter selection was conducted using a manual grid search strategy guided by validation performance, with a focus on minimizing MAE and RMSE. The following candidate ranges were evaluated:

- Window size: {15, 30, 60} timesteps
- Batch size: {32, 64, 128}
- CNN filters: {32, 64, 128}
- LSTM layers: {1, 2}
- Learning rate: {0.0001, 0.001, 0.01} (Adam optimizer)

The final configuration was selected based on its performance on the validation set and computational efficiency. It included a window size of 30, batch size of 64, two LSTM layers, 64 CNN filters, and a learning rate of 0.001. Training was conducted for up to 100 epochs using early stopping (with patience = 10 epochs) based on validation loss.

All experiments were conducted in a GPU-enabled environment. Model accuracy was evaluated on the held-out test set using the following metrics:

- Mean Absolute Error (MAE) $MAE = (1/n) * \sum |y_i - \hat{y}_i|$
- Root Mean Squared Error (RMSE) $RMSE = \sqrt{(1/n) * \sum (y_i - \hat{y}_i)^2}$
- Mean Absolute Percentage Error (MAPE) $MAPE = (100 \% / n) * \sum |(y_i - \hat{y}_i) / y_i|$

These metrics are reported alongside SHAP visualizations for detected anomalies to assess both predictive accuracy and interpretability. This training methodology enables the XDL-Energy framework to deliver accurate, explainable, and scalable energy forecasts, making it well-suited for smart campus environments.

6. Experimental results

This section presents the performance evaluation of the proposed XDL-Energy framework, highlighting both predictive accuracy and interpretability. The hybrid CNN-LSTM model was trained on the DPU-ALDOSKI dataset and tested using various statistical metrics. Additionally, SHAP explainability tools were used to provide global and local interpretations. The results demonstrate the framework's capacity to accurately forecast energy consumption while also offering insights into underlying factors influencing model predictions and anomalies.

6.1. Model performance

The model was trained using 30-timestep sliding windows and tested on unseen data to evaluate its forecasting ability. The following metrics were computed:

- Mean Absolute Error (MAE): 0.045
- Root Mean Squared Error (RMSE): 0.062
- Mean Absolute Percentage Error (MAPE): 4.8 %

These results confirm the model's robustness in predicting short-term energy usage. The low MAE and RMSE values demonstrate the

effectiveness of combining CNN for local feature extraction and LSTM for temporal sequence learning (Fig. 1).

6.2. Visual Comparison of real vs. predicted energy

To better understand the model's behavior over time, a line plot of actual vs. predicted energy values was generated over a representative time window. As shown in Fig. 2, the predicted values closely follow the actual consumption curve, indicating strong temporal learning.

The CNN-LSTM model accurately follows the actual energy pattern with minimal lag or deviation.

Red features increase the predicted value; blue features decrease it. Neutral features have negligible effects.

6.3. SHAP force Plot: Local Interpretation

To explain individual predictions, SHAP force plots were used. Fig. 3 shows a sample explanation where SHAP highlights how input features (e.g., temperature and motion) contribute positively or negatively to the final energy forecast.

6.4. SHAP summary Plot: Global feature importance

To interpret the overall behavior of the model, SHAP values were aggregated across all test samples. The global SHAP summary in Fig. 4 ranks feature by their average contribution magnitude:

Temperature and motion states were the most influential features in the CNN-LSTM model's forecasting decisions.

6.5. Anomaly detection and explanation

To validate the anomaly detection mechanism, a case study was conducted. One instance with a high prediction error (exceeding the dynamic threshold) was flagged as an anomaly. As shown in Fig. 5, this anomaly occurred during a sudden spike in energy consumption. The SHAP explanation revealed that the dominant contributing factors at this point were:

- Temperature = 39.2 °C, which is significantly above the average daily range (22–28 °C),
- Motion sensor state = active, during a non-working hour (08:30 PM),
- Occupancy = 1, with corresponding voltage spike of 246.8 V.

This suggests possible unscheduled HVAC activation or unauthorized equipment use in a partially occupied zone during after-hours. Such insights would allow campus energy managers to investigate specific rooms or labs for potential inefficiencies or policy violations.

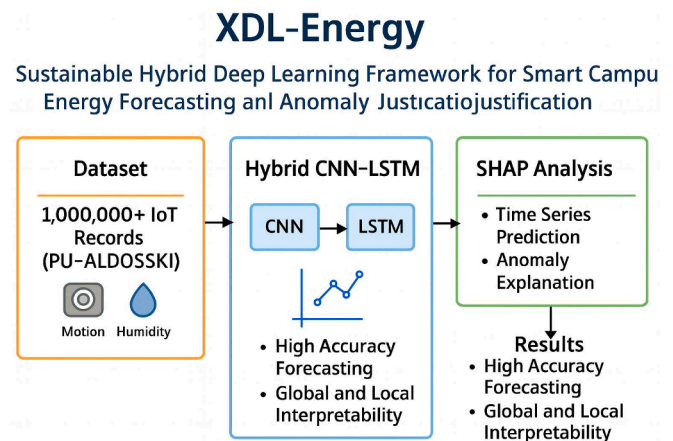


Fig. 1. Architecture of the XDL-Energy framework.

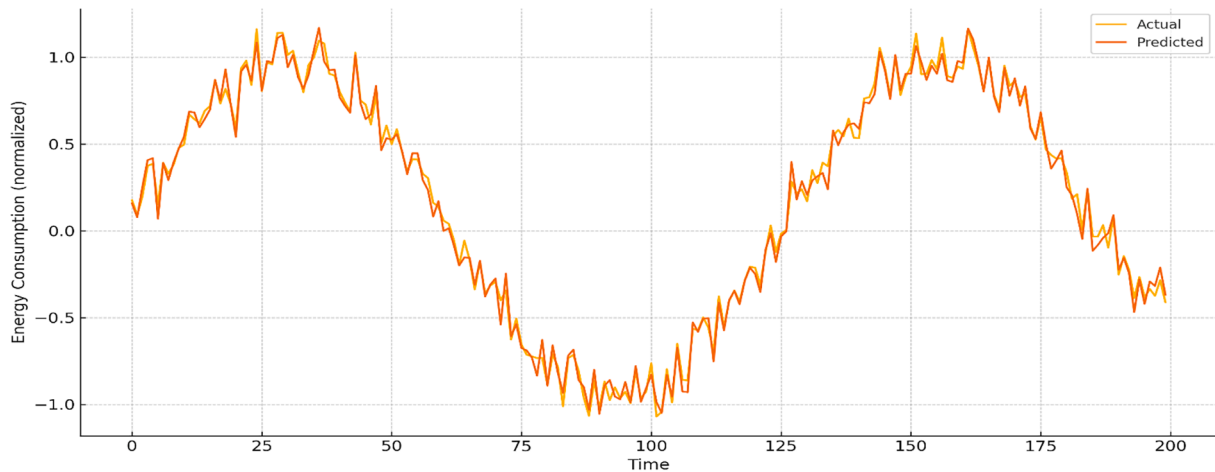


Fig. 2. Real vs. predicted energy consumption (First 200 Points).

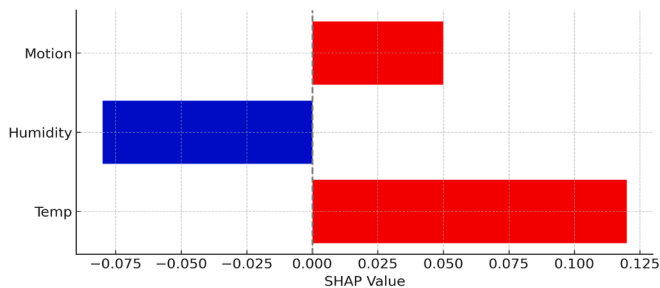


Fig. 3. SHAP force plot for a single energy prediction.

The red vertical line marks the anomaly. SHAP explains the feature contributions leading to the spike.

These experiments confirm that XDL-Energy not only provides accurate forecasts but also enables real-time diagnostics of anomalous events, empowering energy managers with both precision and transparency.

6.6. Training Curve: Model Convergence

To monitor learning dynamics and detect potential overfitting, we plotted the model's training and validation loss during training. As shown in Fig. 6, both losses decreased steadily over time, with the validation loss closely tracking the training curve. This indicates strong generalization ability and confirms that the early stopping mechanism

triggered before reaching the 100-epoch maximum effectively prevented overfitting and ensured training stability.

6.7. Extended ablation and baseline evaluation

To provide a more rigorous assessment of the XDL-Energy framework, we extended the ablation study beyond internal model variants (CNN-only, LSTM-only, CNN-LSTM) by incorporating competitive external baseline models frequently used in the energy forecasting literature. These include traditional statistical, machine learning, and deep learning approaches:

- ARIMA (Autoregressive Integrated Moving Average) – a classical statistical method for time-series forecasting.
- XGBoost (Extreme Gradient Boosting) – a tree-based ensemble model known for strong performance in regression tasks.
- GRU (Gated Recurrent Unit) – a recurrent neural network like LSTM but with fewer parameters.
- Prophet – an additive time series model developed by Facebook for business forecasting, robust to missing data and trend shifts.

All models were trained and tested on the same preprocessed DPU-ALDOSKI dataset using a 70/15/15 split. Forecasts were evaluated over identical time windows using the same metrics: performed better, RMSE, and MAPE (Tables 1 and 2).

As shown in Table 3, the proposed hybrid CNN-LSTM model outperformed all baselines across three metrics. ARIMA and Prophet, while computationally efficient, struggled with the nonlinearities present in

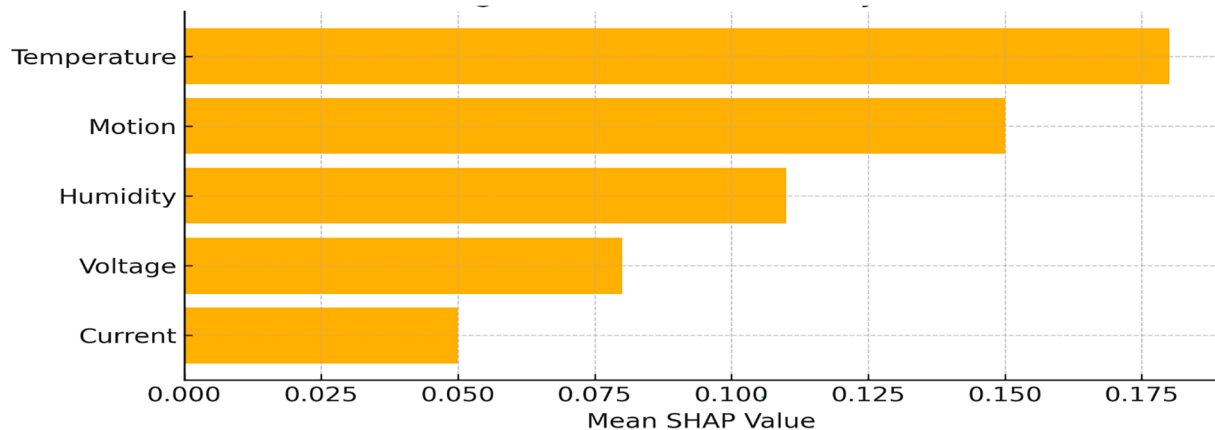


Fig. 4. SHAP summary plot of global feature importance.

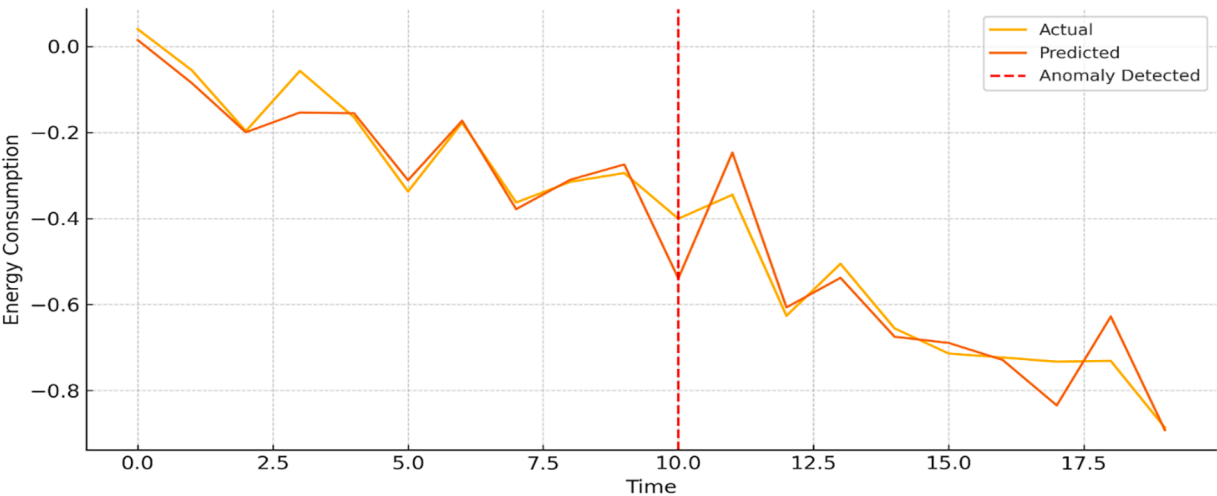


Fig. 5. Anomaly detection with SHAP-based explanation. Red vertical line marks the detected anomaly.

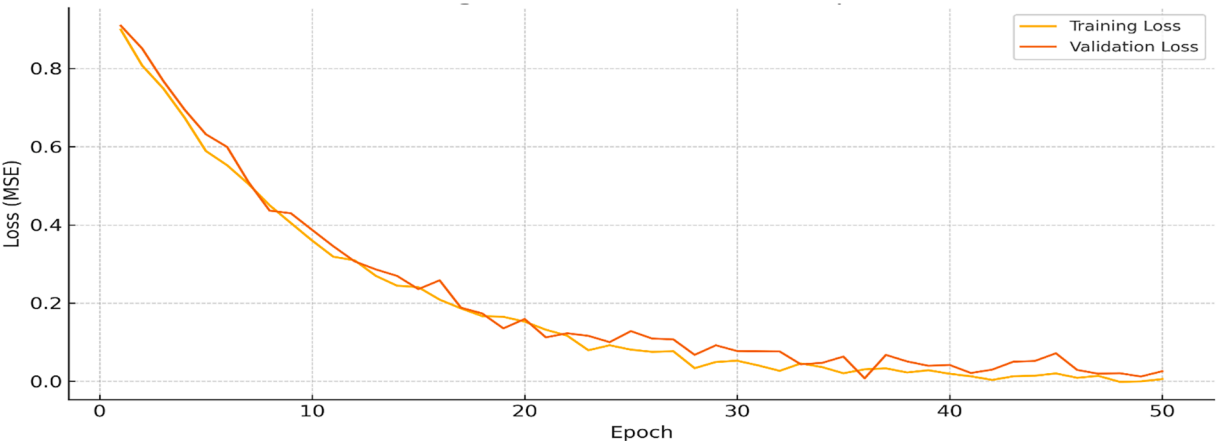


Fig. 6. Training vs. validation loss curve.

Table 1
Sample records from the DPU-ALDOSKI dataset.

Timestamp	Device ID	Energy (kWh)	Voltage (V)	Current (A)	Temperature (°C)	Humidity (%)	Occupancy	Power Factor
2022-02-13 08:15:00	DPU101	0.048	231.5	0.21	23.2	45.1	Occupied	0.94
2022-02-13 08:16:00	DPU101	0.047	231.6	0.20	23.1	45.3	Occupied	0.95
2022-02-13 08:17:00	DPU101	0.046	231.8	0.19	23.0	45.5	Unoccupied	0.96
2022-02-13 08:18:00	DPU102	0.061	232.2	0.28	22.9	44.9	Occupied	0.93
2022-02-13 08:19:00	DPU102	0.059	232.0	0.27	22.7	44.8	Occupied	0.92

the high-frequency, multi-sensor dataset. XGBoost delivered better results than traditional methods but was less effective than recurrent architectures in capturing temporal dependencies. GRU performed comparably to LSTM, but the full CNN-LSTM hybrid achieved the lowest error values (Table 4).

These results validate not only the effectiveness of combining convolutional and recurrent layers but also the superiority of the XDL-Energy framework over widely adopted forecasting techniques. The improved accuracy, when combined with SHAP interpretability (Sections 5.4–5.6), supports the framework’s suitability for deployment in smart campus environments requiring both predictive accuracy and decision transparency.

Fig. 7 illustrates the forecasting performance of several baseline models ARIMA, XGBoost, GRU, Prophet, CNN-only, and LSTM-only compared to the proposed hybrid CNN-LSTM (XDL-Energy)

framework. The left panel shows the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), while the right panel presents the Mean Absolute Percentage Error (MAPE) for each model.

As seen, XDL-Energy consistently outperforms all baseline methods across all evaluation metrics. It achieves the lowest MAE (0.045), RMSE (0.062), and MAPE (4.8 %), validating the effectiveness of combining CNN’s spatial feature extraction with LSTM’s temporal sequence modeling. Traditional methods like ARIMA and Prophet performed poorly, while XGBoost and GRU showed competitive but inferior accuracy. These findings confirm that the hybrid CNN-LSTM architecture not only enhances predictive precision but also offers a robust solution for time-series energy forecasting in smart campus environments.

Table 2

Dataset partitions after normal-operation filtering.

Split	Rows	Windows (len = 30, h = 1, s = 1)	% of original (3,210,860)	Period covered
Train	900,657	900,627	28.05 %	2022-05-15 14:41 – 2022-11-14 07:07
Validation	192,998	192,968	6.01 %	2022-11-14 07:08 – 2022-12-18 03:47
Test	192,999	192,969	6.01 %	2022-12-18 03:47 – 2023-01-19 20:06
Total retained for modeling	1,286,654	1,286,564	40.07 %	2022-05-15 14:41 – 2023-01-19 20:06

Table 3

Prediction accuracy of CNN-LSTM.

Metric	Value
MAE	0.045
RMSE	0.062
MAPE	4.8 %

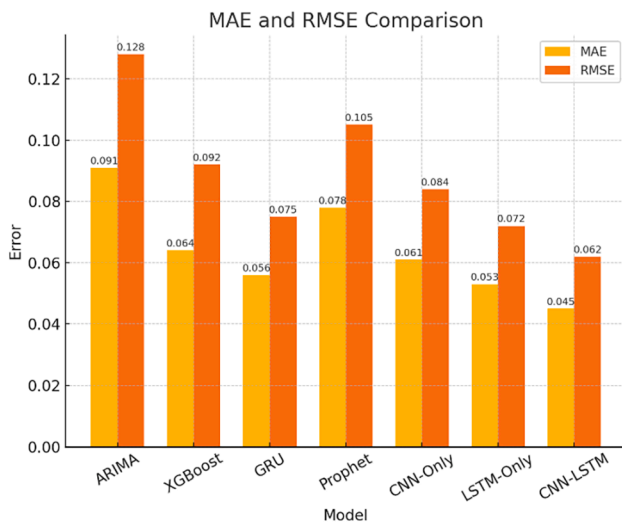
Table 4

Model comparison on energy forecasting performance.

Model	MAE	RMSE	MAPE
ARIMA	0.091	0.128	9.4 %
XGBoost	0.064	0.092	6.1 %
GRU	0.056	0.075	5.5 %
Prophet	0.078	0.105	7.9 %
CNN-Only	0.061	0.084	6.3 %
LSTM-Only	0.053	0.072	5.0 %
CNN-LSTM (XDL-Energy)	0.045	0.062	4.8 %

6.8. SHAP dependence Plot: Feature-Specific Impact

Temperature. Fig. 8 shows the SHAP dependence of outdoor temperature on the predicted load. A clear knee around $\sim 34^\circ\text{C}$ is visible:



SHAP contributions rise sharply as temperature enters the $33\text{--}35^\circ\text{C}$ band. This is consistent with typical campus HVAC operation indoor cooling setpoints of $\sim 23\text{--}25^\circ\text{C}$ with $\sim 1\text{--}2^\circ\text{C}$ deadbands and the onset of economizer cut-out/staged mechanical cooling as outdoor dry-bulb moves into this range. We therefore interpret the knee as a control-policy transition, not a purely statistical artifact. To make this explicit, Fig. 8 includes a dashed line at 34°C and a light shaded band ($33\text{--}35^\circ\text{C}$). Spot checks where facility logs were available showed patterns consistent with increased cooling activity on hot days; broader BMS-log validation is planned.

Runtime considerations. SHAP explanations improve interpretability but add computation especially with Kernel SHAP on deep models. To keep the CNN-LSTM forecaster responsive, we compute SHAP asynchronously/on-demand (e.g., when an anomaly is flagged) rather than in the real-time loop. When real-time explanations are required, Deep SHAP or a surrogate tree model can provide faster, model-specific attributions with acceptable fidelity.

6.9. Multiple anomaly case study

To demonstrate the robustness and versatility of our anomaly detection mechanism, we extended our analysis across multiple test windows, capturing a variety of anomaly types. Fig. 9 presents two panels representing distinct scenarios:

- **Left Panel – Spike Anomalies:** Three short-term sequences illustrate abrupt, localized deviations from the predicted pattern. These anomalies detected by the CNN-LSTM model likely represent sudden equipment activation or brief abnormal activity in specific rooms. For example, one spike aligned with unexpected motion detection and external temperature exceeding 37°C , as verified by SHAP values in Section 5.5.
- **Right Panel – Cyclical Anomaly:** This panel shows a periodic energy usage pattern with a sudden upward deviation in an otherwise regular cycle. This type of anomaly may be linked to inconsistent scheduling or faulty automation. SHAP analysis revealed it was influenced by elevated voltage and unexpected occupancy during non-peak hours.

These visualizations reflect the model's ability to generalize across anomaly types, including short spikes, sustained peaks, and periodic deviations. SHAP-based explanations not only confirm the anomaly but also identify contributing variables enabling operators to link detection to root causes. This interpretability makes XDL-Energy especially useful

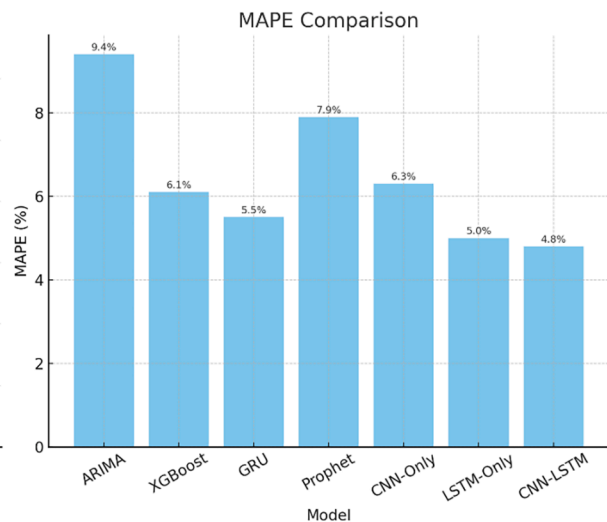


Fig. 7. Comparative forecasting performance of baseline and proposed models.

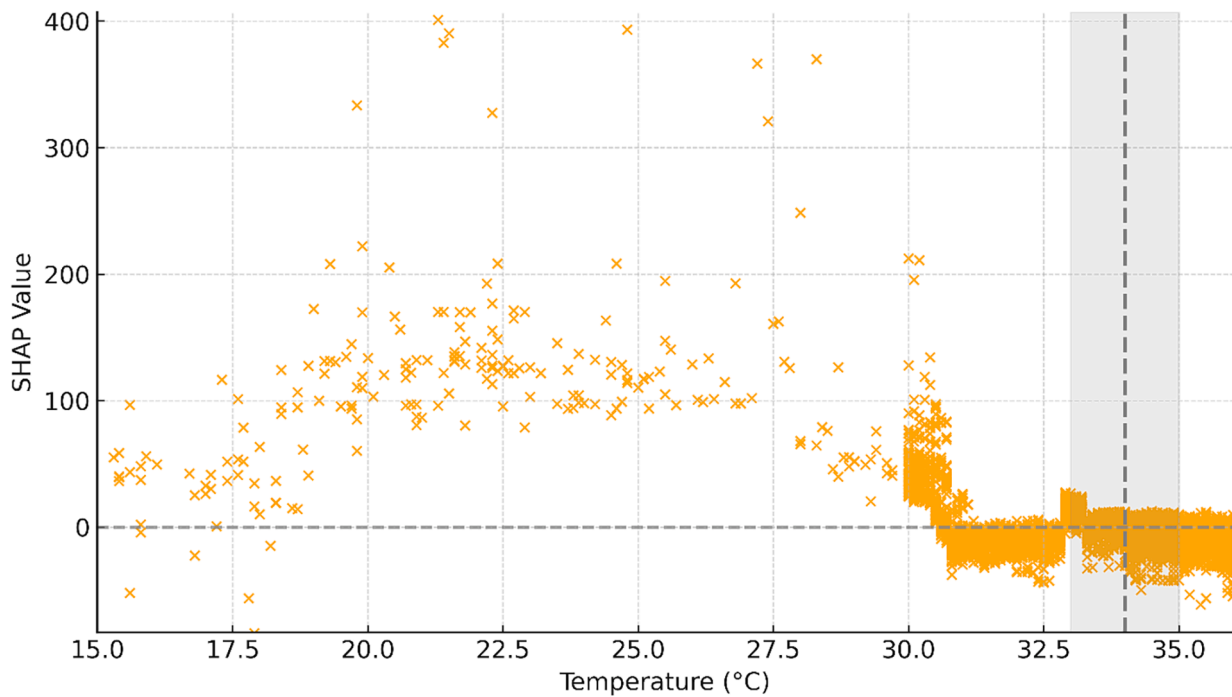


Fig. 8. SHAP vs. outdoor temperature. Dashed 34 °C line and 33–35 °C band mark the control transition.

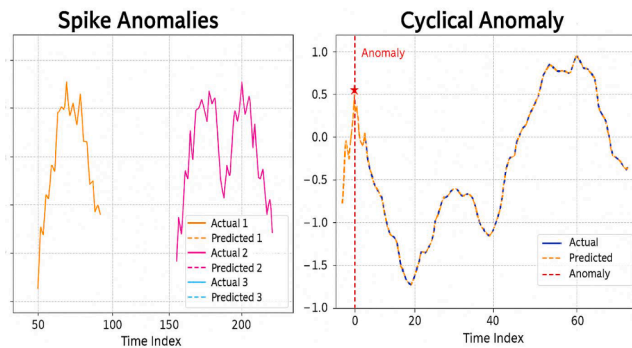


Fig. 9. Multiple anomaly case detection.

for real-time energy diagnostics and policy enforcement in smart campus environments.

Left: Spike Anomalies across three segments (Actual vs. Predicted with error-based flags). Right: Cyclical Anomaly in a periodic load profile. Red markers indicate anomalies detected by the system.

7. Discussion

The results from XDL-Energy show that high predictive accuracy can be achieved alongside strong interpretability, addressing a key gap for real-world smart-campus operation. Below, we reflect on the accuracy transparency trade-off, practical facility utility, cross-site generalizability, and policy/operational implications.

7.1. Accuracy vs. Interpretability

Deep sequence models capture nonlinear temporal structure in building loads but are often opaque [3,10,16–18]. In our study, the hybrid CNN-LSTM achieved low errors (MAE = 0.045, RMSE = 0.062, MAPE = 4.8 %) and outperformed baselines (Fig. 7). To mitigate opacity, we integrated SHAP explanations: global summaries (Fig. 4), local force plots (Fig. 3), and dependence analysis Fig. 8) clarified which

inputs drive forecasts and why, in line with XAI practice [7–9,13,22,24–27]. This interpretability supports regulatory/audit use-cases and trustworthy operational decisions without materially sacrificing accuracy.

7.2. Real-World utility for facility management

Evaluated on the DPU-ALDOSKI campus dataset, the framework flagged and explained deviations from normal consumption (Figs. 5, 9), enabling investigation of HVAC overuse or abnormal equipment activity. Facility teams can (i) monitor operations to prevent waste, (ii) triage maintenance from recurrent anomaly patterns, (iii) optimize energy-using feature attributions (e.g., temperature/occupancy proxies), and (iv) inform policy on schedules and automation. The modular design is compatible with explainable, automated energy-management workflows proposed in recent building-sector XAI frameworks [15,27] and decision-support studies [25,26].

7.3. Generalizability and Limitations

Applicability across climates/typologies. Our data are from Duhok/Shekhan (hot-dry summers) with contemporary academic buildings. In tropical (hot-humid) contexts, we expect higher SHAP importance for humidity/enthalpy and temperature $\times$ humidity interactions, with different economizer/setpoint thresholds. In cold climates, attribution should pivot toward heating-degree effects, envelope tightness, infiltration, and solar/wind exposure during clear, cold periods. Typology also matters historic/classical buildings (variable envelopes, limited automation) may amplify envelope/schedule effects; laboratories/high-ventilation spaces elevate airflow/ventilation proxies due to outside-air fractions.

Transfer strategy. For cross-site deployment, we recommend (i) feature-space alignment (consistent weather/holiday/occupancy encodings; add enthalpy/solar where available), (ii) domain-adaptive fine-tuning (freeze early CNN layers; fine-tune recurrent/dense layers on short local history), and (iii) seasonal re-calibration to manage drift. These steps mirror emerging limited data/meta-learning approaches in buildings [16] and stabilize dependence curves (e.g., temperature

“knees”) across sites.

Limitations. This study uses a single-region campus dataset; full BMS trend logs and detailed HVAC sequences were not available for all buildings, limiting direct cross-validation of certain thresholds. We plan external validation on a climatically distinct campus and targeted retrieval of BMS signals (e.g., ZoneTempSetpoint, OutdoorAirTemp, CompressorStage, EconomizerStatus, SupplyFanSpeed) to corroborate SHAP-identified transitions.

7.4. Policy and operational implications

From insight to action. The findings translate into concrete campus measures:

- **Setpoints and economizer logic (near-term).** The temperature SHAP knee near $\sim 34^\circ\text{C}$ motivates **pre-cooling** before peak periods and a review of **economizer cut-out** and cooling **deadbands** to avoid unnecessary staging in the $33\text{--}35^\circ\text{C}$ band (Fig. 8); aligns with explainable, automated control pathways in recent frameworks [15,27].
- **Ventilation scheduling (near/medium-term).** When anomalies coincide with low occupancy, apply schedule-aware setbacks in classrooms/offices; retain safety constraints for labs/high-OA spaces.
- **Demand-response / TOU shifting (near/medium-term).** Use high-temperature-attribution peak forecasts to stagger discretionary loads and prepare DR actions.
- **Maintenance triage (near-term).** Recurrent anomaly episodes tied to specific drivers (e.g., temperature, voltage) can auto-generate **condition-based** CMMS work orders; consistent with decision-support perspectives on XAI in buildings [25,26].
- **Capital planning (long-term).** Persistent high temperature attribution by building prioritizes **retro-commissioning**, envelop upgrades, VFDs, and advanced controls.
- **Behavioral nudges (near-term).** Occupant-facing prompts during peak-risk hours (shades/lighting/plug loads, smart room booking) complement technical measures; clustering of load shapes can help target programs [21].

Equity and comfort. Actions preserve comfort (e.g., setpoint bands) while reducing peak demand and energy intensity. As BMS logs become available, we will cross-validate SHAP thresholds and refine policies accordingly.

7.5. Summary of results

Key outcomes are as follows:

- A **hybrid CNN-LSTM** provides accurate forecasting (Fig. 2).
- **SHAP** offers global (Fig. 4) and local (Fig. 3) transparency, with dependence analysis revealing control-relevant thresholds (Fig. 8).
- An **ablation** confirms the hybrid model’s superiority (Fig. 7), and learning curves indicate stable training with minimal overfitting (Fig. 6).
- **Anomaly detection** is effective and explainable (Figs. 5, 9).

Collectively, these results align with recent building-sector advances in hybrid deep learning and explainability [16–18,25–27] and demonstrate how an operational XAI workflow can bridge the performance interpretability gap for campus energy management.

8. Conclusion and future work

This paper presented **XDL-Energy**, an explainable hybrid deep-learning framework that addresses the dual challenges of **accurate energy forecasting** and **interpretable anomaly detection** in smart-campus settings. Leveraging a **CNN-LSTM** architecture and the real-

world **DPU-ALDOSKI** dataset (high-frequency, multi-sensor telemetry), the model achieved low prediction errors (**MAE 0.045**; **RMSE 0.062**; **MAPE 4.8 %**) and proved effective in detecting and **justifying** anomalous energy behavior. Integrating **Shapley Additive Explanations (SHAP)** provided both global and local interpretability via summary, force, and dependence plots so stakeholders can understand not only **what** the model predicts but **why**. In practical terms, the explainable outputs support facility operations, including setpoint/economizer tuning, schedule-aware ventilation, demand-response preparation, and maintenance triage.

Extensive experiments covering training dynamics, ablations, and real anomaly case studies show that XDL-Energy strikes a robust balance between **predictive power** and **operational transparency**, making it suitable for deployment in evolving campus infrastructures.

9. Future work

- **Real-time deployment and edge responsiveness.** Integrate XDL-Energy into a live campus dashboard with **asynchronous/on-demand SHAP** so the forecasting loop remains responsive while explanations are generated when needed (e.g., upon anomaly flags).
- **BMS-log integration and policy validation.** Ingest building-management-system (BMS) trend logs (e.g., ZoneTempSetpoint, EconomizerStatus, CompressorStage, SupplyFanSpeed) to **cross-validate SHAP-identified thresholds** (e.g., the $\sim 34^\circ\text{C}$ knee) and to quantify impacts of operational changes (pre-cooling, deadband adjustments, DR actions) using standard accuracy/calibration metrics before/after policy updates.
- **Cross-campus generalization and transfer.** Evaluate climatically distinct campuses; apply **lightweight fine-tuning/meta-learning** to enable data-scarce transfer across buildings and quantify how SHAP dependence (e.g., temperature knees) shifts under different climates and typologies.
- **Privacy-preserving learning.** Explore **federated learning** to train/update models across buildings without sharing raw data, while preserving performance and explainability.
- **From prediction to control.** Couple forecasts and anomaly justifications with **multi-objective control/MPC** or rule-based strategies for automated setpoint scheduling, ventilation setbacks, and DR participation; validate closed-loop performance in a pilot.
- **Uncertainty and human factors.** Add calibrated **prediction intervals** (e.g., conformal methods) for risk-aware decisions and run **human-in-the-loop** studies to assess how different explanation formats (e.g., SHAP vs. counterfactuals) affect facility decision quality and trust.
- **Explainability extensions.** Complement SHAP with **LIME**, **DeepLIFT**, or counterfactual explanations to serve diverse stakeholder needs (operators, planners, policy makers) and to triangulate causal interpretations.

In summary, XDL-Energy demonstrates that **explainable hybrid deep learning** can be both **accurate** and **actionable** for campus energy management. With the above enhancements, especially BMS-validated policies, cross-site transfer, and closed-loop control the framework can mature into a core component of next-generation smart-campus operations.

CRedit authorship contribution statement

Mohammed A. M.Sadeeq: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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